Controls on the Distribution of Extensional Strain Between Rifting Events

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The Rifting Cycle: Three Phases

Continental rifting proceeds in three phases:

1. **Co-rifting** — near-instantaneous elastic strain release via dike emplacement and normal faulting
2. **Post-rifting** — visco-elastic relaxation decaying over years to decades
3. **Inter-rifting** — secular plate divergence between events

Problem: the mechanisms that build up strain *between* events remain the least constrained part of the cycle.

Key observations

- Iceland & Afar episodes geodetically documented
- Post-diking transients decay within years
- Far-field velocities must converge to plate rate $\dot{u}_\infty$

The Kinematic Gap

Standard "backslip" models treat the rift as a dislocation opening continuously in an elastic half-space (Savage & Prescott 1978).

Physical failures:

- Ignores weak, viscous lower crust & upper mantle
- Infinitely deep dike is mechanically untenable
- Short relaxation times in magmatically active regions are ignored
- Finite numerical domains force $u = 0$ at boundaries, incompatible with observed far-field divergence

The gap

No model simultaneously reproduces the *transient* post-diking signal *and* the *steady-state* far-field plate divergence.

We bridge this gap with a fully analytical, cyclic model.

Physical Model: Elastic Crust over Viscous Half-Space

Setup: elastic plate of thickness H (brittle crust) coupled to a Newtonian viscous half-space (asthenosphere, viscosity η).

Stress balance (depth-averaged):

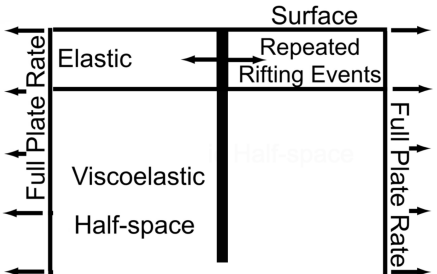
$$H\,\frac{\partial \bar{\sigma}_{xx}}{\partial x} + \sigma_{xz}^{\mathrm{BOT}} = 0$$

Basal shear via the non-local **Stokes kernel**:

$$\sigma_{xz}^{\mathrm{BOT}}(x,t) = \int_{-\infty}^{+\infty} \frac{2\eta}{\pi (x-x')^2}\, \dot{u}(x',t)\, dx'$$

Key parameters

H	elastic plate thickness
E	Young's modulus
ν	Poisson's ratio
η	asthenospheric viscosity
ω	half-opening per dike event
T	recurrence interval



Fractional Diffusion and the Characteristic Velocity

Fourier transform of the governing integro-differential equation yields a simple ODE for each wavenumber k:

$$\frac{\partial \hat{u}}{\partial t} = -a\,|k|\,\hat{u} \qquad \Longleftrightarrow \qquad \frac{\partial u}{\partial t} = -a\,(-\Delta)^{1/2}\,u$$

The **characteristic velocity** a encodes the elastic–viscous coupling:

$$\boxed{a := \frac{EH}{2\eta(1-\nu^2)}}$$

Physical meaning

a sets the *speed* at which post-diking strain redistributes away from the rift axis. Large a (stiff crust, low viscosity) $\Rightarrow$ fast relaxation; small a $\Rightarrow$ slow, broad deformation pattern.

Single-Event Step Response

After one dike intrusion (Heaviside IC), the displacement and velocity are:

$$u^{\mathrm{STEP}}(x,t) = \frac{\omega}{\pi}\arctan\left(\frac{at}{x}\right)$$

$$v^{\mathrm{STEP}}(x,t) = \frac{\omega a}{\pi}\,\frac{x}{x^2 + a^2 t^2}$$

- Displacement grows as arctan — logarithmically slow at large x
- Velocity has a Lorentzian shape, widening in time
- **Far-field:** $u \to \omega/2$ as $t \to \infty$ (not zero!)

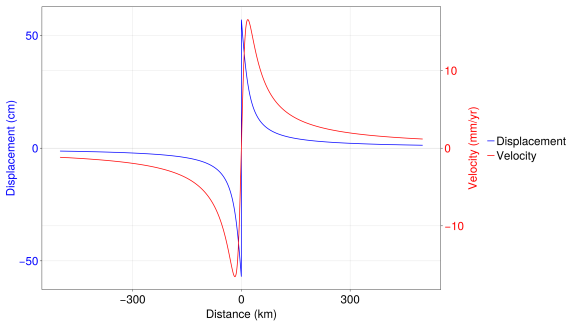


Figure: CSS best-fit displacement (blue) and velocity (red) profiles.

Cyclo-Steady State (CSS): Summing an Infinite History

After many cycles the system reaches a **periodic steady state**. Summing the contributions of all past intrusions using the **Digamma** (ψ) and **log-Gamma** ($\ln\Gamma$) functions:

$$v_{\mathrm{CSS}}(x,\tau) = \frac{2\omega}{\pi T}\,\mathrm{Im}\Big\{\psi\Big(\frac{\tau}{T} + i\frac{x}{aT}\Big)\Big\}$$

$$u_{\mathrm{CSS}}(x,\tau) = \frac{2\omega}{\pi}\,\mathrm{Im}\Big\{\ln\Gamma\Big(\frac{\tau}{T} + i\frac{x}{aT}\Big) - \ln\Gamma\Big(i\frac{x}{aT}\Big)\Big\}$$

where $\tau \in [0, T)$ is the time elapsed since the last intrusion.

Key property: mass conservation

$$\int_0^T v_{\mathrm{CSS}}(x,\tau)\,d\tau = \omega, \quad \forall\, x > 0$$

The entire right half-plate moves by ω per cycle — consistent with global plate motion models.

Maxwell Visco-Elastic Asthenosphere

Upgrade: replace Newtonian fluid with a Maxwell body (spring + dashpot in series), relaxation time $\tau_R = \eta/\mu_m$.

Modified evolution equation (Fourier domain):

$$\frac{\partial \hat{u}}{\partial t} = -\frac{a\,|k|}{1 + \tau_R a\,|k|}\,\hat{u}$$

- High-k (near-field): decays at fixed rate $1/\tau_R$ (elastic phase)
- Low-k (far-field): recovers pure viscous $\Rightarrow$ same CSS limit

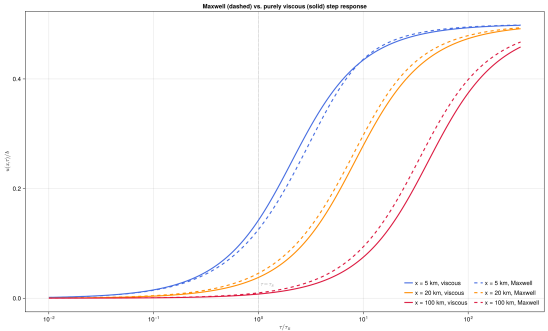


Figure: Maxwell (dashed) vs. purely viscous (solid) step response. Near the rift ($x = 5$ km) the Maxwell model is initially slower, converging at $\tau/\tau_R \gg 1$.

Model Comparison: 1-Parameter Fits

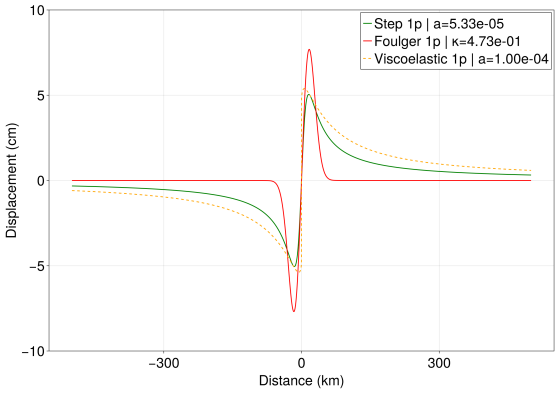


Figure: **Displacement.** Step (green) and Viscoelastic (orange) match the broad antisymmetric pattern; Foulger kernel (red) overshoots.

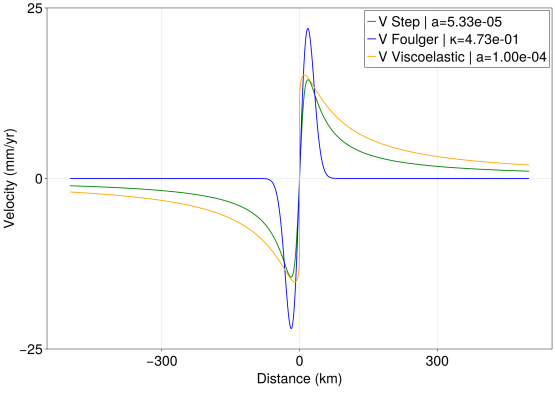


Figure: **Velocity.** The purely elastic Foulger kernel (blue) produces an unphysical spike; both fractional-diffusion models yield physically smooth profiles.

Fitting Foulger et al. (1992) Iceland GPS Data

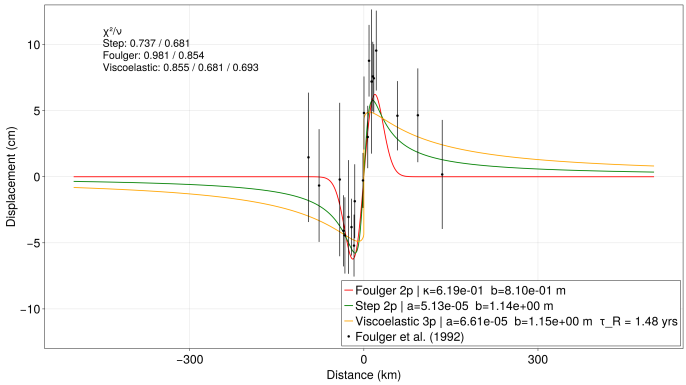


Figure: 2- and 3-parameter fits to post-diking displacement. Step ($\chi^2/\nu = 0.737$) and Viscoelastic ($\chi^2/\nu = 0.855$) both beat the Foulger kernel ($\chi^2/\nu = 0.981$).

Best-fit parameters

Model	a (m/s)
Step 2p	4.79×10^{-5}
Viscoelastic	6.69×10^{-5}
	τ_R (yr)
Viscoelastic	1.87

The CSS step model achieves the *best* fit with only 2 free parameters.

Constraining τ_R and Physical Parameters

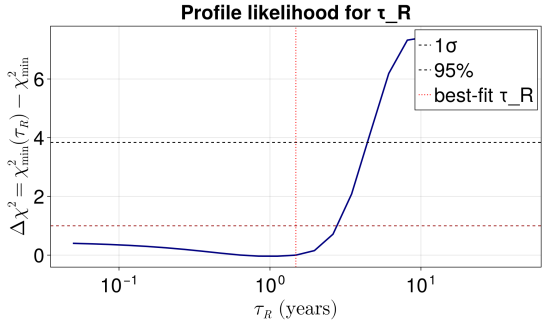


Figure: Profile likelihood for τ_R. Best fit: $\tau_R \approx 1.87$ yr. Values > 5 yr excluded at 95% confidence.

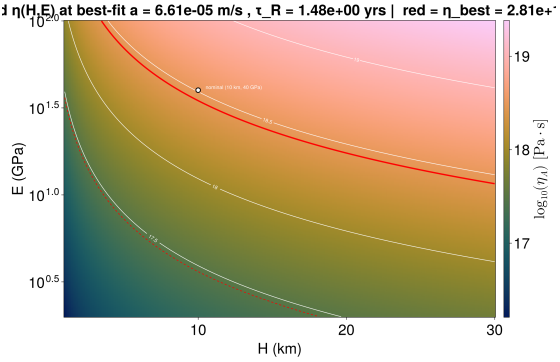


Figure: Implied $\eta(H, E)$ at best-fit a and τ_R. Canonical lithosphere ($H = 10$ km, $E = 40$ GPa) implies $\eta \approx 3.5 \times 10^{18}$ Pa·s.

Summary & Conclusions

1. We derive a **fully analytical, cyclic model** for inter-rifting strain based on fractional (half-Laplacian) diffusion.
2. The **CSS solution** (Digamma / log-Gamma functions) guarantees consistency with far-field plate motion by construction.
3. A single parameter $a = EH/[2\eta(1-\nu^2)]$ controls the spatial breadth of deformation.
4. **Fits to Iceland GPS** (Foulger 1992) yield $a \approx 4.8 \times 10^{-5}$ m/s and, in the Maxwell model, $\tau_R \approx 1.9$ yr $\Rightarrow \eta \sim 3.5 \times 10^{18}$ Pa·s.
5. The CSS step model **outperforms** classical diffusion kernels without requiring unphysical far-field boundary conditions.

Take-home message

Inter-rifting deformation is the *superposition* of viscous responses from an infinite history of discrete dike events. The "steady" interseismic signal is not steady at all — it is the emergent, periodic sum of transients.